DO NOW — RECALL

Normalisation to 3NF and indexing

Name: _____ Class: _____ Date: _____

1. What is a transitive dependency?

2. What does an index help with?

3. What does 1NF require?

4. Simplify (A∧B)∨(A∧¬B), stating the Boolean law.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An attribute depends on key through another non-key attribute.

2. Model answer

Search speed on indexed fields.

3. Model answer

Atomic fields and no repeating groups.

4. Model answer

A∧(B∨¬B)=A∧1=A.

Vocabulary — know these terms

1NF — Atomic values/no repeating groups.

2NF — No partial dependency on part of composite key.

3NF — No non-key transitive dependency violating key dependency.

index — Search structure over columns.

NEW LEARNING

Read, annotate and explain

Understand

1NF requires atomic values and no repeating groups in a record; 2NF removes dependencies on only part of a composite candidate key.

Apply

3NF eliminates inappropriate transitive dependencies among non-key facts, reducing repeated customer names across orders.

Evaluate

An index accelerates some lookups but uses storage and may slow insert/update operations. Normalisation and indexing solve different problems.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A table Order(OrderID,CustomerID,CustomerName) repeats a name. Show a 3NF decomposition.

Predict first

Store customer name once in Customer and reference by CustomerID in Order.

Model answer

Explain the reasoning and justify the output.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State 1NF,2NF,3NF in one sentence each. [6]

Question 2

Order(OrderID, CustomerID, CustomerName). Explain the redundancy and split into two tables. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why an index on StudentName can speed SELECT but also incur costs. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Normalising to 3NF means the database never needs indexes.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.